FRI Natural language processing course 2026
Final report

University of Ljubljana
Faculty of Computer and Information Science

A RAG-Based Slovenian Legal Assistant for Real-Estate Law

Max Lazar, Tilen Vodička, Nik Kosednar

Abstract

General purpose large language models often lack reliable knowledge of specific legal systems and may hallucinate unsupported legal claims. In this project, we build a retrieval augmented generation (RAG) system for Slovenian legal question answering over real-estate and property-law legislation from the PISRS dataset. Legal texts are split into article level chunks, embedded with BGE-M3, stored in a FAISS index, and answered by GaMS3-12B-Instruct. We evaluate retrieval and answer quality on a few question sets, testing three GaMS model sizes, different embedding models, BM25 and hybrid retrieval, MMR, re-ranking, query expansion and generation parameter sweeps.

Keywords

retrieval-augmented generation, Slovenian law, NLP, PISRS, legal question answering, FAISS, BM25, GaMS

1. Introduction

Legal question answering is a knowledge intensive task, answers must not only sound plausible, but must be supported by the correct legal provisions. This is especially important in the legal domain, where unsupported or outdated claims can be misleading. Our goal was to build a Slovenian legal assistant that answers questions about real-estate and property-law legislation. The implemented system retrieves relevant legal sources and generates short answers.

We focus on retrieval augmented generation (RAG), where a language model is not expected to answer purely from its internal parameters. Instead, a retriever first selects relevant legal text passages and the generator answers based on those passages. This follows the general RAG idea introduced by Lewis et al. [1]. We use FAISS for vector similarity search [2], BGE-M3 as the main multilingual embedding model [3] and Slovenian GaMS models [4] as the generator. We also test multilingual-E5-large [5] and Slovlo-v1 [6].

2. Methods

2.1 Corpus and chunking

We used the PISRS legal documents from the COLESLAW PISRS data, especially `register-predpisov.jsonl`. We filtered the corpus with keywords related to real-estate such as *nepremičnine*, *zemljiška knjiga*, *hipoteka*, *gradbeno dovoljenje*, and *urejanje prostora*. Documents were split into article level chunks using article headings such as *170. člen*. The chunking script also normalized Unicode, whitespace, punctuation, warning lines, and structural heading artifacts. Our final data and index statistics can be seen in Table 1.

Table 1. Final data and index statistics.

Item	Value
PISRS records read	7,375
Filtered legal documents kept	3,468
Final article chunks	78,904
Limited early chunks	23,592
Final FAISS index size	309 MB
Final metadata size	157 MB
Development evaluation set	30 questions
Final article-derived set	50 questions

2.2 RAG pipeline and models

The final pipeline has four steps. First, the user question is embedded. Second, FAISS retrieves the most similar legal article chunks. Third, the top contexts are inserted into a legal QA prompt and fourth, the GaMS model generates a short answer grounded in the retrieved text.

The final selected configuration was:

- Generator: `cjvt/GaMS3-12B-Instruct`
- Embedding: `BAAI/bge-m3`
- Retrieval: dense FAISS retrieval
- Reranker: not used in final setup
- MMR: not used in final setup

- Prompt: short extractive prompt (P2)
- Context chunks: `context_n = 2`
- Context length: 1200 characters per chunk
- Generation budget: 220 new tokens

3. Evaluation Setup

We used two main evaluation sets. The 30 question development set contained realistic user style questions: easy, hard, true/false, comparison, scenario, and tricky negative questions. It was useful for development, but some gold sources were ambiguous. The final dataset was generated with LLM assistance from randomly sampled PISRS article chunks, which we fed to ChatGPT Plus. For each sampled chunk, the LLM generated a realistic legal question and an expected answer hint based on the provided article text. Because the source chunk was passed to the model for question generation, we considered it as the gold article for retrieval evaluation. This reduced ambiguity compared with manually labeling realistic user questions, where several different sources may be valid. For the final 50 question article derived set, we generated 40 single source, and 10 multi source questions, to evaluate how the model performs if there are many different sources. The question types are defined as seen in Table 2.

Table 2. Different question difficulties and type categories.

Category	Description
Easy	Direct factual or definitional questions, usually answerable from one article.
Hard	Single source questions requiring careful reading, conditions, exceptions, or legal procedure.
Comparison	Questions comparing two concepts, records, or procedures.
Scenario	Practical situations asking which rule applies.
Yes/No	Questions requiring a yes/no answer and justification.
Tricky negative	Questions with false assumptions, negation, or overgeneralization.
Multi source	Questions requiring evidence from two or more articles.

Retrieval was evaluated with Top-1, Top-3, Top-5, and Top-10 accuracy/recall against the gold article(s). For multi source questions, we measured both whether at least one required source was retrieved and whether all required sources were retrieved. Answer quality was evaluated as *correct*, *partial*, or *wrong*. Answer evaluation was done manually and using LLM as a judge using the expected (gold) source articles, retrieved chunks, answer hint and generated answer.

4. Experiments and Results

Altogether we tested limited vs. full corpus indexing, improved chunk cleanup, three GaMS model sizes, no-RAG vs. RAG, five prompt variants, output cleaning, different context and generation parameter sweep, BGE reranking, MMR, query expansion, three embedding models, BM25 and hybrid BM25 with dense retrieval and multi source context size. In the end we also ran the model on a general/unanswerable question set, to see how the model responds when it does not have enough information to answer the question.

4.1 RAG, model size, and prompts

We first compared no-RAG and RAG versions of GaMS-2B, GaMS-9B, and GaMS3-12B on the 30 question development set, to see if RAG offers any improvement compared to no-RAG setups. RAG improved all model sizes substantially (Figure 1). Across all three models, no-RAG achieved 23/90 correct, 41/90 partial, and 26/90 wrong answers, while RAG achieved 63/90 correct, 20/90 partial, and only 7/90 wrong answers. This confirms that grounding the model in retrieved PISRS sources is more important than relying on parametric knowledge alone.

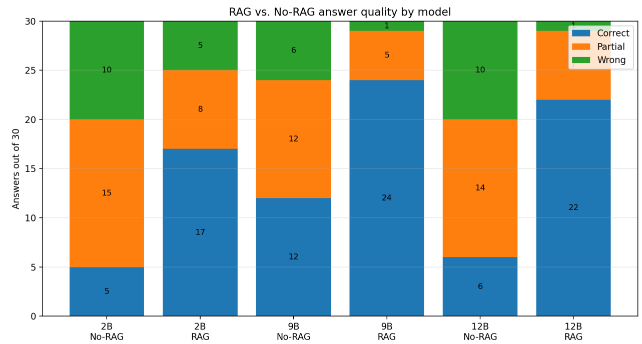

Figure 1. Comparison of no-RAG vs RAG for GaMS 2B, 9B and 12B models on the 30-question development set.

The best result on this set was GaMS-9B with RAG (24/30 correct), though we selected GaMS3-12B for final experiments because it produced cleaner and more complete answers in qualitative runs. We also tested five prompt variants and selected the short extractive prompt P2, because it produced the fewest formatting artifacts. We added output cleanup: special token skipping, stop marker cleanup, markdown cleanup, and invalid citation removal. After cleanup, remaining errors were mostly legal correctness and source completeness, not formatting.

4.2 Embedding and retrieval experiments

Figure 2 shows the embedding comparison on the final 50-question set. BGE-M3 gave the best recall and was kept as the baseline.

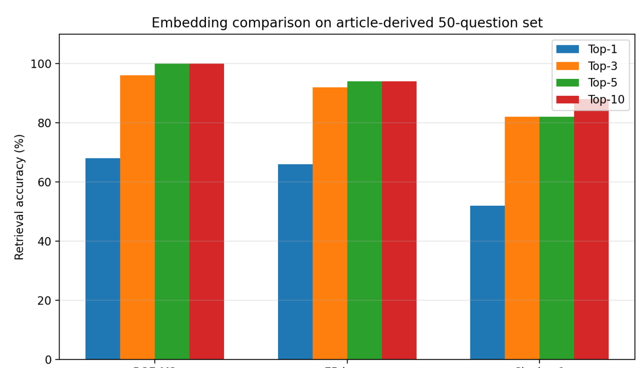

Figure 2. Embedding comparison on the final 50 questions dataset.

We tried several retrieval improvements. Query expansion was mixed. On the 30-question development set it improved Top-1 from 66.67% to 76.67% and Top-3 from 86.67% to 93.33%. However, on the final article-derived set it reduced single-source Top-1 from 77.5% to 67.5% and Top-5 from 100% to 97.5%. We therefore did not use query expansion in the final system. A BGE re-ranker did not reliably improve Top-1 and sometimes promoted semantically similar but legally inappropriate chunks. MMR also hurt recall: for BGE-M3, Top-3 dropped from 96% to 80% and Top-5 from 100% to 92% (Table 3). We therefore did not use re-ranking or MMR in the final dense setup.

Emb.	Mode	T1	T3	T5	T10
BGE	Dense	68%	96%	100%	100%
BGE	MMR	68%	80%	92%	100%
E5	Dense	66%	92%	94%	94%
E5	MMR	66%	70%	74%	88%
Slovlo	Dense	52%	82%	82%	88%
Slovlo	MMR	52%	64%	68%	82%

Table 3. Dense retrieval vs. MMR on the final dataset. MMR used candidate $k = 30$, final $k = 10$, $\lambda = 0.7$.

We also implemented BM25 and hybrid BM25 with dense retrieval. BM25 alone was worse than dense retrieval, but hybrid retrieval with $\alpha = 0.75$ strongly improved Top-1 from 68% to 86% on the 50 question set, while slightly reducing Top-3/Top-5 recall (Table 4). This shows that exact keyword matching helps promote correct legal articles, while dense retrieval remains better for high recall.

Method	T1	T3	T5	T10
Dense BGE-M3	68%	**96%**	**100%**	**100%**
BM25	54%	66%	68%	78%
Hybrid $\alpha = 0.25$	62%	76%	84%	94%
Hybrid $\alpha = 0.50$	70%	88%	94%	98%
Hybrid $\alpha = 0.75$	**86%**	90%	98%	**100%**

Table 4. BM25 and hybrid retrieval on the final 50-question set.

4.3 Generation settings

We swept four context/generation settings: 700/160, 700/220, 1200/160 and 1200/220, where the first number is context characters per chunk and the second is generated tokens. The 1200/220 setting had no cutoff endings and the fewest formatting problems, so it was used for final generation. Increasing context length does not change retrieval, it only gives the model more text from retrieved articles.

5. Final Evaluation

Table 5 shows final dense retrieval. The system achieved 100% Top-5 recall when at least one gold source was required. Single source questions were strong, but multi source questions remained difficult when all required sources had to be retrieved.

Hybrid $\alpha = 0.75$ improved answer correctness, correct answers increased from 29/50 to 34/50, while partial answers decreased from 15/50 to 10/50. The number of wrong answers stayed the same. Hybrid helped single-source ranking, but still did not solve multi-source (2-3 sources) coverage.

Set	T1	T3	T5	T10
All, at least one	68%	96%	100%	100%
Single source	77.5%	97.5%	100%	100%
Multi source, at least one	30%	90%	100%	100%
Multi source, all sources	0%	40%	50%	60%

Table 5. Final dense retrieval evaluation.

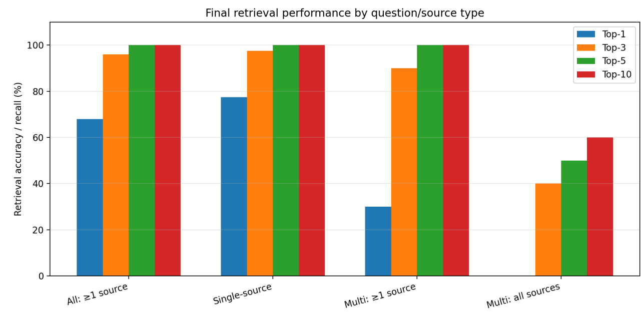

Figure 3. Final retrieval performance split by source type.

For the dense final system, 58% of answers were correct, 30% partial, and 12% wrong (Table 6). The largest weakness was multi source answering with only 1/10 multi source answers being fully correct, although 7/10 were partially correct.

Type	Correct	Partial	Wrong
Easy	8/10	2/10	0/10
Hard	7/10	2/10	1/10
Comparison	3/5	2/5	0/5
Scenario	4/5	0/5	1/5
Yes/No	3/5	0/5	2/5
Tricky neg.	3/5	2/5	0/5
Multi-source	1/10	7/10	2/10

Table 6. Final answer quality by question type.

Additionally, we tested 10 general and 10 unanswerable questions, to see how the model answers when there are no sources available. The model handled many broad legal questions reasonably, but unanswerable questions were risky. It sometimes correctly refused, but other times hallucinated concrete facts, such as inventing a notary for a specific address. Some example inputs and outputs showing correct grounding, correct refusal and hallucination on unanswerable questions can be seen in Table 7.

Question	Observed answer behavior
Kaj pomeni služnost?	Correctly answered from SPZ: service/easement is the right to use another's thing or require omission of acts.
Kdo je danes lastnik parcele 123/4?	Correctly refused to answer: the provided legal sources do not contain current ownership data.
Kateri notar je overil pogodbo za Celovško 55?	Failure case: hallucinated a concrete notary, showing weak answerability detection.

Table 7. Example outputs showing correct grounding, correct refusal, and hallucination on unanswerable questions.

6. Discussion

RAG substantially improved grounding compared with no-RAG generation. Across GaMS-2B, GaMS-9B, and GaMS3-12B, wrong answers decreased from 26/90 to 7/90, while correct answers increased from 23/90 to 63/90. This confirms that for legal QA, retrieved legal sources are more important than relying on the model's parametric knowledge alone.

The main bottleneck was retrieval and source visibility. Dense BGE-M3 achieved strong Top-5 recall, but the generator only sees the selected context chunks. With `context_n=2`, if the gold article is ranked third or lower, the model cannot use it and may answer from a related article, prior knowledge, or guess. This explains why high retrieval recall does not always translate into fully correct answers.

Retrieval method also mattered. Slovlo-v1 performed worse than BGE-M3, suggesting that retrieval training quality matters more than Slovenian-specific pretraining alone. MMR and reranking did not help in our setup. Hybrid BM25 with dense retrieval was more useful: $\alpha = 0.75$ improved Top-1 from 68% to 86% and increased fully correct answers from 29/50 to 34/50, showing that exact legal terminology and semantic similarity are complementary.

Multi-source and unanswerable questions remained the main weaknesses. Multi-source questions often require more than two chunks, while increasing `context_n` to 5 also increased verbosity and citation risk. For unanswerable questions, the model sometimes correctly refused, but also sometimes hallucinated concrete facts. Future systems should therefore use adaptive context size, query decomposition, and stronger answerability detection.

7. Future Work

The most direct improvement is adaptive retrieval. Use context_n=2 for simple questions and more chunks for multi source or comparison questions. Another improvement is query decomposition, where a complex legal question is split into sub queries and evidence is retrieved separately for each part, which is especially relevant for multi source questions. Additional (larger) models could be tested in the future. Other future work includes conversational RAG where follow up questions are rewritten into standalone questions before retrieval. If local RAG context is insufficient, future systems could fall back to trusted official sources such as PISRS or court/authority websites, for legal QA, but unrestricted web search would be unsafe and could provide false information.

8. Conclusion

We built a complete Slovenian legal RAG system over PISRS legal documents. The final dense baseline achieved 100% Top-5 retrieval recall on the 50 question article derived dataset, with 58% fully correct answers and 30% partially correct answers. The main lesson is that legal RAG quality depends more on source selection and evidence coverage than on model size alone.

References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

[2] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

[3] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, 2024.

[4] CJVT. GaMS3-12B-Instruct. https://huggingface.co/cjvt/GaMS3-12B-Instruct, 2024.

[5] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Multilingual E5 text embeddings: A technical report. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 2089–2103, 2024.

[6] CJVT. SloVlo-v1: Slovenian text embeddings. https://huggingface.co/cjvt/slovlo-v1, 2024.